

KERALA DIPLOMA CURRICULUM REVISION 2026

Course 5123A

5 Credits: 4

Program Elective

Source-grounded

Statistical Quality Assurance

□ To impart knowledge on Quality, Quality Control And Assurance

OFFICIAL SOURCE DECLARATION

How this lesson was prepared

The supplied official SITTTTR syllabus PDF for Course 5123A is the authoritative source for course information, objectives, course outcomes, CO-PO mapping, modules, topics, activities, assessment structure and references. Explanations, study prompts and Malayalam support are educational elaboration and are not presented as additional official syllabus text.

Source file: 5123A.pdf from the uploaded official SITTTTR syllabus archive.

Course code and title: preserved from the source.

COURSE DASHBOARD

Official course information

Course information

Item	Official information
Programme	Diploma in Wood and Paper Technology
Course code	5123A
Course title	Statistical Quality Assurance
Semester	5 Credits: 4
Course category	Program Elective
Periods per week	4 (L:4, T:0, P:0) Periods per semester: 60
Periods per semester	60
Credits	4
Prerequisite	See the official syllabus PDF
Assessment	Use the official assessment section reproduced below

Modules

4 official module sections detected.

Topic entries

15 source-aligned topic entries retained.

Study mode

Theory and application emphasis.

STUDY METHOD**How to use this lesson****First pass**

Read the official source declaration, dashboard and module transcript. Mark unfamiliar terms without changing the official terminology.

Second pass

Use each topic card to build a definition, principle, steps or components, application and exam answer. Attempt the Q&A before opening the answers.

Practical evidence

Where the course is laboratory or workshop based, maintain an observation notebook with procedure, instruments, readings, safety points and conclusion.

Revision pass

Return to the source excerpt, coverage table, activities and model paper. The generated paper is practice material, not an official examination paper.

LEARNING OUTCOMES

Objectives, outcomes and mapping

Course objectives

- To impart knowledge on Quality, Quality Control And Assurance
- To introduce the concept of Statistical Quality Control.
- To illustrate the Control Chart.
- To design acceptance Sampling and O C Curve.
- Course-prerequisites:
- Knowledge of basic Mathematics Mathematics I 1
- Knowledge in Statistics Mathematics II 2

Course outcomes

CO	Official outcome	Study level
CO1	15 Understanding Assurance	See official syllabus
CO2	Analyze the Statistical quality control 14 Applying	See official syllabus
CO3	Prepare the control chart 15 Applying	See official syllabus
CO4	Solve acceptance sampling and o c curve. 14 Applying	See official syllabus

CO-PO mapping evidence

Row	Extracted official mapping
CO1	CO1
CO2	CO2 3
CO3	CO3 3
CO4	CO4 3

COVERAGE AUDIT

Complete syllabus coverage

Every detected official module is represented below. Each module contains topic cards and an expandable official syllabus transcript to preserve source terminology.

Module coverage

Module	Official heading	Topic entries	Hours
Module 1	Outline the Quality, Quality Control And Assurance	4	As specified
Module 2	Analyze the Statistical quality control	4	As specified
Module 3	Prepare the control chart	3	As specified
Module 4	Solve acceptance sampling and o c curve.	4	As specified

MODULE 1

Outline the Quality, Quality Control And Assurance

This module is presented from the official syllabus scope for Statistical Quality Assurance. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 1 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Outline the Quality, Quality Control And Assurance
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

▼ Explain quality

Explain quality is an official topic in Module 1 of Statistical Quality Assurance. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain quality using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain quality should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain quality means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-എന്നു പറയുന്നതുകൊണ്ട് നോക്കി നിശ്ചയിക്കുക. നിശ്ചയിക്കുക എന്നതിന്റെ, steps, application എന്നിവയെക്കുറിച്ചും നോക്കുക. Technical terms English-ൽ എഴുതുക. നിശ്ചയിക്കുക.

▼ Discuss quality assurance system

Discuss quality assurance system is an official topic in Module 1 of Statistical Quality Assurance. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Discuss quality assurance system using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, discuss quality assurance system should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Discuss quality assurance system means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1- Discuss quality assurance system
 പദപരിചയം definition/meaning. പദപരിചയം, പദപരിചയം
 പദപരിചയം, steps, application പദപരിചയം പദപരിചയം പദപരിചയം. Technical terms
 English- പദപരിചയം പദപരിചയം.

▼ Explain quality assurance program.

Explain quality assurance program. is an official topic in Module 1 of Statistical Quality Assurance. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain quality assurance program. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain quality assurance program. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain quality assurance program. means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1- Explain quality assurance program.
 പദപരിചയം definition/meaning.
 പദപരിചയം, steps, application
 പദപരിചയം. Technical terms
 English- പദപരിചയം.

▼ Summarize quality in materials management 4 Understanding Quality assurance. Quality - Quality Control - Total Quality Control-Quality Assurance - Forms of Quality Assurance - stages - Quality Planning-Quality Assurance Programme - Objectives - Quality in materials management.

Summarize quality in materials management 4 Understanding Quality assurance. Quality - Quality Control - Total Quality Control-Quality Assurance - Forms of Quality Assurance - stages - Quality Planning-Quality Assurance Programme - Objectives - Quality in materials management. is an official topic in Module 1 of Statistical Quality Assurance. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Summarize quality in materials management 4 Understanding Quality assurance. Quality - Quality Control - Total Quality Control-Quality Assurance - Forms of Quality Assurance - stages - Quality Planning-Quality Assurance Programme - Objectives - Quality in materials management. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, summarize quality in materials management 4 understanding quality assurance. quality - quality control - total quality control- quality assurance - forms of quality assurance - stages - quality planning-quality assurance programme - objectives - quality in materials management. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Summarize quality in materials management 4 Understanding Quality assurance. Quality - Quality Control - Total Quality Control- Quality Assurance - Forms of Quality Assurance - stages - Quality Planning-Quality Assurance Programme - Objectives - Quality in materials management. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1- Summarize quality in materials management 4 Understanding Quality assurance. Quality - Quality Control - Total Quality Control-Quality Assurance - Forms of Quality Assurance - stages - Quality Planning-Quality Assurance Programme - Objectives - Quality in materials management. definition/meaning . steps, application . Technical terms English- .

► **Official syllabus detail and terminology (expand in digital version)**

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 2

Analyze the Statistical quality control

This module is presented from the official syllabus scope for Statistical Quality Assurance. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 2 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Analyze the Statistical quality control
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

▼ Explain statistical quality chart

Explain statistical quality chart is an official topic in Module 2 of Statistical Quality Assurance. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain statistical quality chart using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain statistical quality chart should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain statistical quality chart means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2- Explain statistical quality chart
 പദപരിചയം definition/meaning. പദപരിചയം, steps, application പദപരിചയം പദപരിചയം. Technical terms English- പദപരിചയം പദപരിചയം.

▼ Develop frequency program.

Develop frequency program. is an official topic in Module 2 of Statistical Quality Assurance. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Develop frequency program. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, develop frequency program. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Develop frequency program. means in the official course context
Study lens	Principle → components or stages → result → application

Definition, labelled explanation, comparison, procedure or example as appropriate

000000000000 00000 definition/meaning 000000000000. 000000 000000
 00000000, steps, application 000000 0000000000 000000. Technical terms
 English-0 00000 000000000000.

▼ Calculate mean ,mode and median

Calculate mean ,mode and median is an official topic in Module 2 of Statistical Quality Assurance. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Calculate mean ,mode and median using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, calculate mean ,mode and median should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Calculate mean ,mode and median means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2- കണക്കാക്കുക, മോഡ്, മീഡിയൻ
 നിർവ്വചനം/അർത്ഥം, ഉദാഹരണം. ഏകദേശം, ഏകദേശം
 ഏകദേശം, steps, application ഉദാഹരണം ഉദാഹരണം. Technical terms
 English-ൽ ഉദാഹരണം ഉദാഹരണം.

▼ Apply regression analysis 3 Applying Statistical Quality Control.

Definition of SQC - Characteristics - attributes and Variables - process of SQC Frequency distribution - Frequency Polygon - Frequency Curve - Measure of central Tendency - Mean - Median - mode - Probability - Normal curve - Regression analysis - Normal Test-x2 Test - Comparison between 100% inspection and sampling Inspection.

Apply regression analysis 3 Applying Statistical Quality Control. Definition of SQC - Characteristics - attributes and Variables - process of SQC Frequency distribution - Frequency Polygon - Frequency Curve - Measure of central Tendency - Mean - Median - mode - Probability - Normal curve - Regression analysis - Normal Test-x2 Test - Comparison between 100% inspection and sampling Inspection. is an official topic in Module 2 of Statistical Quality Assurance. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Apply regression analysis 3 Applying Statistical Quality Control. Definition of SQC - Characteristics - attributes and Variables - process of SQC Frequency distribution - Frequency Polygon - Frequency Curve - Measure of central Tendency - Mean - Median - mode - Probability - Normal curve - Regression analysis - Normal Test-x2 Test - Comparison between 100% inspection and sampling Inspection. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.

- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, apply regression analysis 3 applying statistical quality control. definition of sqc - characteristics - attributes and variables - process of sqc frequency distribution - frequency polygon - frequency curve - measure of central tendency - mean - median - mode - probability - normal curve - regression analysis - normal test-x2 test - comparison between 100% inspection and sampling inspection. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Apply regression analysis 3 Applying Statistical Quality Control. Definition of SQC - Characteristics - attributes and Variables - process of SQC Frequency distribution - Frequency Polygon - Frequency Curve - Measure of central Tendency - Mean - Median - mode - Probability - Normal curve - Regression analysis - Normal Test-x2 Test - Comparison between 100% inspection and sampling Inspection. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2- Apply regression analysis 3 Applying Statistical Quality Control. Definition of SQC - Characteristics - attributes and Variables - process of SQC Frequency distribution - Frequency Polygon - Frequency Curve - Measure of central Tendency - Mean - Median - mode - Probability - Normal curve - Regression analysis - Normal Test-x2 Test - Comparison between 100% inspection and sampling Inspection.

മലയാള സാധനങ്ങൾക്ക് definition/meaning നൽകുന്നു. മലയാള സാധനങ്ങൾക്ക് steps, application നൽകുന്നു. Technical terms English-ൽ നൽകുന്നു.

► **Official syllabus detail and terminology (expand in digital version)**

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 3

Prepare the control chart

This module is presented from the official syllabus scope for Statistical Quality Assurance. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 3 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Prepare the control chart
- Official duration: As specified
- Official topic entries captured: 3
- The full source excerpt is preserved below for coverage checking.

▼ Explain variable control chart

Explain variable control chart is an official topic in Module 3 of Statistical Quality Assurance. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain variable control chart using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain variable control chart should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain variable control chart means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3- Explain variable control chart

വിശദീകരിക്കുക definition/meaning വിശദീകരിക്കുക. വിശദീകരിക്കുക വിശദീകരിക്കുക, steps, application വിശദീകരിക്കുക വിശദീകരിക്കുക. Technical terms English- വിശദീകരിക്കുക വിശദീകരിക്കുക.

▼ Develop different chart in SQC

Develop different chart in SQC is an official topic in Module 3 of Statistical Quality Assurance. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Develop different chart in SQC using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, develop different chart in sqc should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Develop different chart in SQC means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-ാം ഡെവലപ്പ് ചെയ്ത ചാർട്ട് SQC

ചാർട്ട്/ചാർട്ടിംഗ് ഫോർമ് definition/meaning ചാർട്ട്/ചാർട്ടിംഗ് ഫോർമ്. ചാർട്ട്/ചാർട്ടിംഗ് ഫോർമ്, steps, application ചാർട്ട്/ചാർട്ടിംഗ് ഫോർമ്. Technical terms English-ൽ ചാർട്ട്/ചാർട്ടിംഗ് ഫോർമ്.

▼ Develop solution based on P chart/ U Chart 5 Applying Control Charts Control charts for Variables - X and R charts - calculation of 3σ Limits - Construction of X and r Charts - Moving Range charts - Accuracy and Precision - P chart - Pn/nP chart/U chart

Develop solution based on P chart/ U Chart 5 Applying Control Charts
Control charts for Variables - X and R charts - calculation of 3σ Limits -
Construction of X and r Charts - Moving Range charts - Accuracy and
Precision - P chart - Pn/nP chart/U chart is an official topic in Module 3 of
Statistical Quality Assurance. Study the terminology first, then connect the
stated purpose, sequence, components or application to the wider course
outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Develop solution based on P chart/ U Chart 5 Applying Control Charts Control charts for Variables - X and R charts - calculation of 3σ Limits - Construction of X and r Charts - Moving Range charts - Accuracy and Precision - P chart - Pn/nP chart/U chart using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, develop solution based on p chart/ u chart 5 applying control charts control charts for variables - x and r charts - calculation of 3σ limits - construction of x and r charts - moving range charts - accuracy and precision - p chart - pn/np chart/u chart should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Develop solution based on P chart/ U Chart 5 Applying Control Charts Control charts for Variables - X and R charts - calculation of 3σ Limits - Construction of X and r Charts - Moving Range charts - Accuracy and Precision - P chart - Pn/nP chart/U chart means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3- Develop solution based on P chart/ U Chart 5 Applying Control Charts Control charts for Variables - X and R charts - calculation of 3σ Limits - Construction of X and r Charts - Moving Range charts - Accuracy and Precision - P chart - Pn/nP chart/U chart definition/meaning . steps, application . Technical terms English- .

► **Official syllabus detail and terminology (expand in digital version)**

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 4

Solve acceptance sampling and o c curve.

This module is presented from the official syllabus scope for Statistical Quality Assurance. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 4 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Solve acceptance sampling and o c curve.
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

▼ Explain sampling 3 Remembering Develop the difference between sampling and

Explain sampling 3 Remembering Develop the difference between sampling and is an official topic in Module 4 of Statistical Quality Assurance. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain sampling 3 Remembering Develop the difference between sampling and using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain sampling 3 remembering develop the difference between sampling and should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
--------	-----------------

Malayalam support: Module 4- Explain sampling 3 Remembering
Develop the difference between sampling and population definition/meaning. Explain the steps, application of sampling. Technical terms English- Explain in Malayalam.

▼ 4 Applying control chart Distinguish single sampling and double

4 Applying control chart Distinguish single sampling and double is an official topic in Module 4 of Statistical Quality Assurance. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Applying control chart Distinguish single sampling and double using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 applying control chart distinguish single sampling and double should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 4 Applying control chart Distinguish single sampling and double means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4- 4 Applying control chart Distinguish single sampling and double sampling definition/meaning. steps, application. Technical terms English- Malayalam.

▼ 3 Applying sampling plan

3 Applying sampling plan is an official topic in Module 4 of Statistical Quality Assurance. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Applying sampling plan using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 applying sampling plan should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Applying sampling plan means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-പാഠ്യ 3 Applying sampling plan പരീക്ഷിക്കുക
 പരീക്ഷിക്കുക definition/meaning പരീക്ഷിക്കുക. പരീക്ഷിക്കുക പരീക്ഷിക്കുക പരീക്ഷിക്കുക, steps,
 application പരീക്ഷിക്കുക പരീക്ഷിക്കുക പരീക്ഷിക്കുക. Technical terms English-പാഠ്യ പരീക്ഷിക്കുക
 പരീക്ഷിക്കുക.

▼ Apply the problems on reliability 4 Applying Acceptance Sampling And OC Curve. Acceptance Sampling - Principle of Lot sampling - Limitations of sampling - Comparison between Acceptance sampling and control Charts - Benefits - Single sampling plan double sampling Plan - OC Curve - Lot Tolerance Percent Defective(LTPD) - AOQ, Reliability - Simple Problems Text /Reference: T/R Book Title/Author T1 M.Mahajan. Published by Dhanpatharai & company R2 Industrial Engg and management by O.P. Khanna R3 M.Mahajan. Statistical Quality control ,S Chand R4 K.C.Aroa,.Total Quality Management ,Dhanapat Online Resources: SI.No Website Link 1 <https://www.youtube.com/watch?v=CfCztL5DkTE> 2 <https://www.youtube.com/watch?v=qb3mvJ1gb9g> 3 <https://www.youtube.com/watch?v=QsLDBfApqol> 4 <https://www.youtube.com/watch?v=e5g2NmlUdck> 5 https://www.youtube.com/watch?v=GC6k_qbpwmA

Apply the problems on reliability 4 Applying Acceptance Sampling And OC Curve. Acceptance Sampling - Principle of Lot sampling - Limitations of sampling - Comparison between Acceptance sampling and control Charts - Benefits - Single sampling plan double sampling Plan - OC Curve - Lot Tolerance Percent Defective(LTPD) - AOQ, Reliability - Simple Problems Text /Reference: T/R Book Title/Author T1 M.Mahajan. Published by Dhanpatharai & company R2 Industrial Engg and management by O.P. Khanna R3 M.Mahajan. Statistical Quality control ,S Chand R4 K.C.Aroa,.Total Quality Management ,Dhanapat Online Resources: SI.No Website Link 1 <https://www.youtube.com/watch?v=CfCztL5DkTE> 2 <https://www.youtube.com/watch?v=qb3mvJ1gb9g> 3 <https://www.youtube.com/watch?v=QsLDBfApqol> 4 <https://www.youtube.com/watch?v=e5g2NmlUdck> 5 https://www.youtube.com/watch?v=GC6k_qbpwmA is an official topic in Module 4 of Statistical Quality Assurance. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Apply the problems on reliability 4 Applying Acceptance Sampling And OC Curve. Acceptance Sampling - Principle of Lot sampling - Limitations of sampling - Comparison between Acceptance sampling and control Charts – Benefits - Single sampling plan double sampling Plan - OC Curve - Lot Tolerance Percent Defective(LTPD) - AOQ, Reliability - Simple Problems Text /Reference: T/R Book Title/Author T1 M.Mahajan. Published by Dhanpatharai & company R2 Industrial Engg and management by O.P. Khanna R3 M.Mahajan. Statistical Quality control ,S Chand R4 K.C.Aroa,.Total Quality Management ,Dhanapat Online Resources: Sl.No Website Link 1 <https://www.youtube.com/watch?v=CfCztL5DkTE> 2 <https://www.youtube.com/watch?v=qb3mvJ1gb9g> 3 <https://www.youtube.com/watch?v=QsLDBfApqol> 4 <https://www.youtube.com/watch?v=e5g2NmlUdck> 5 https://www.youtube.com/watch?v=GC6k_qbpwMA using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, apply the problems on reliability 4 applying acceptance sampling and oc curve. acceptance sampling - principle of lot sampling - limitations of sampling - comparison between acceptance sampling and control charts – benefits - single sampling plan double sampling plan - oc curve - lot tolerance percent defective(ltpd) - aoq, reliability - simple problems text

/reference: t/r book title/author t1 m.mahajan. published by dhanpatharai & company r2 industrial engg and management by o.p. khanna r3 m.mahajan. statistical quality control ,s chand r4 k.c.aroa,.total quality management ,dhanapat online resources: sl.no website link 1 <https://www.youtube.com/watch?v=cfcztL5dkTE> 2 <https://www.youtube.com/watch?v=qb3mvj1gb9g> 3 <https://www.youtube.com/watch?v=QsLDBfApqoI> 4 <https://www.youtube.com/watch?v=e5g2NmlUdck> 5 https://www.youtube.com/watch?v=GC6k_qbpwmA should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Apply the problems on reliability 4 Applying Acceptance Sampling And OC Curve. Acceptance Sampling - Principle of Lot sampling - Limitations of sampling - Comparison between Acceptance sampling and control Charts - Benefits - Single sampling plan double sampling Plan - OC Curve - Lot Tolerance Percent Defective(LTPD) - AOQ, Reliability - Simple Problems Text /Reference: T/R Book Title/Author T1 M.Mahajan. Published by Dhanpatharai & company R2 Industrial Engg and management by O.P. Khanna R3 M.Mahajan. Statistical Quality control ,S Chand R4 K.C.Aroa,.Total Quality Management ,Dhanapat Online Resources: Sl.No Website Link 1 https://www.youtube.com/watch?v=CfCztL5DKTE 2 https://www.youtube.com/watch?v=qb3mvj1gb9g 3 https://www.youtube.com/watch?v=QsLDBfApqoI 4 https://www.youtube.com/watch?v=e5g2NmlUdck 5 https://www.youtube.com/watch?v=GC6k_qbpwmA means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Malayalam support: Module 4-□□ Apply the problems on reliability 4
Applying Acceptance Sampling And OC Curve. Acceptance Sampling -
Principle of Lot sampling - Limitations of sampling - Comparison between
Acceptance sampling and control Charts - Benefits - Single sampling plan
double sampling Plan - OC Curve - Lot Tolerance Percent Defective(LTPD) -
AOQ, Reliability - Simple Problems Text /Reference: T/R Book Title/Author T1
M.Mahajan. Published by Dhanpatharai & company R2 Industrial Engg and
management by O.P. Khanna R3 M.Mahajan. Statistical Quality control ,S
Chand R4 K.C.Aroa,.Total Quality Management ,Dhanapat Online Resources:
Sl.No Website Link 1 <https://www.youtube.com/watch?v=CfCztL5DkTE> 2
<https://www.youtube.com/watch?v=qb3mvJ1gb9g> 3
<https://www.youtube.com/watch?v=QsLDBfApqol> 4
<https://www.youtube.com/watch?v=e5g2NmlUdck> 5
https://www.youtube.com/watch?v=GC6k_qbpwmA □□□□□□□□□□ □□□□□
definition/meaning □□□□□□□□□□. □□□□□ □□□□□ □□□□□□, steps,
application □□□□□ □□□□□□□□ □□□□□. Technical terms English-□ □□□□□
□□□□□□□□□□.

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

RAPID REFERENCE**Concept and formula bank****Answer structure**

Definition → Principle →
Components/steps →
Application → Conclusion

Use this sequence for descriptive questions when the topic does not require a numerical formula.

Practical record

Aim → Apparatus → Procedure
→ Observation → Result →
Safety

Use this sequence for laboratory, workshop and field activities.

RAPID REVISION**Diagram and source-transcript library****Module 1 source and topic cards**

Jump to official terminology, topic explanations and study guidance.

Module 2 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 3 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 4 source and topic cards

Jump to official terminology, topic explanations and study guidance.

OFFICIAL SELF-LEARNING**Activities from the syllabus**

Use the extracted official activity wording as the record of required self-learning work.

The official PDF does not expose a separate activities heading in the extracted text; consult the source PDF pages for the exact activity list.

EXAMINATION PREPARATION

Assessment methodology

The following source extract preserves the assessment information detected in the official PDF. Verify mark conversions and institutional updates against the current source before an examination.

The official assessment structure is reproduced from the supplied PDF.

Practice-paper notice: The model paper below is generated for revision and is not an official examination paper.

ACTIVE RECALL

Module-wise questions and answers

These are practice questions based on official topic coverage, not official examination questions.

Module 1

► Define or explain Explain quality. (3 marks)

► Define or explain Discuss quality assurance system. (3 marks)

► Define or explain Explain quality assurance program.. (3 marks)

► Define or explain Summarize quality in materials management 4 Understanding Quality assurance. Quality - Quality Control - Total Quality Control-Quality Assurance - Forms of Quality Assurance - stages - Quality Planning-Quality Assurance Programme - Objectives - Quality in materials management.. (3 marks)

Module 2

► Define or explain Explain statistical quality chart. (3 marks)

► Define or explain Develop frequency program.. (3 marks)

► Define or explain Calculate mean ,mode and median. (3 marks)

► Define or explain Apply regression analysis 3 Applying Statistical Quality Control. Definition of SQC - Characteristics - attributes and Variables - process of SQC Frequency distribution - Frequency Polygon - Frequency Curve - Measure of central Tendency - Mean - Median - mode

- Probability - Normal curve - Regression analysis - Normal Test-x2 Test - Comparison between 100% inspection and sampling Inspection.. (3 marks)

Module 3

► **Define or explain Explain variable control chart. (3 marks)**

► **Define or explain Develop different chart in SQC. (3 marks)**

► **Define or explain Develop solution based on P chart/ U Chart 5 Applying Control Charts Control charts for Variables - X and R charts - calculation of 3σ Limits - Construction of X and r Charts - Moving Range charts - Accuracy and Precision - P chart - Pn/nP chart/U chart. (3 marks)**

Module 4

► **Define or explain Explain sampling 3 Remembering Develop the difference between sampling and. (3 marks)**

► **Define or explain 4 Applying control chart Distinguish single sampling and double. (3 marks)**

► **Define or explain 3 Applying sampling plan. (3 marks)**

► **Define or explain Apply the problems on reliability 4 Applying Acceptance Sampling And OC Curve. Acceptance Sampling - Principle of Lot sampling - Limitations of sampling - Comparison between Acceptance sampling and control Charts - Benefits - Single sampling plan double sampling Plan - OC Curve - Lot Tolerance Percent Defective(LTPD) - AOQ, Reliability - Simple Problems Text /Reference: T/R Book Title/Author T1 M.Mahajan. Published by Dhanpatharai & company R2 Industrial Engg and management by O.P. Khanna R3**

M.Mahajan. Statistical Quality control ,S Chand R4 K.C.Aroa,.Total Quality Management ,Dhanapat Online Resources: Sl.No Website Link

1	https://www.youtube.com/watch?v=CfCztL5DkTE
2	https://www.youtube.com/watch?v=qb3mvJ1gb9g
3	https://www.youtube.com/watch?v=QsLDBfApqol
4	https://www.youtube.com/watch?v=e5g2NmIUdck
5	https://www.youtube.com/watch?v=GC6k_qbpwmA. (3 marks)

PRACTICE ONLY

Practice Model Paper — Not an Official Examination Paper

Attempt one short definition, one structured explanation and one long answer from every module. Use the official assessment pattern reproduced above when selecting time and marks.

Module 1

1-mark definition: State the meaning of **Explain quality**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 2

1-mark definition: State the meaning of **Explain statistical quality chart**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 3

1-mark definition: State the meaning of **Explain variable control chart**.

3-mark explanation: Explain one principle, process or comparison from this module.

Module 4

1-mark definition: State the meaning of **Explain sampling 3 Remembering Develop the difference between sampling and**.

3-mark explanation: Explain one principle, process or comparison from

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

QUICK REVISION

Exam checklist

Before writing

- Read the command word: define, explain, compare, draw, calculate, analyse or design.
- Use the exact course terminology from the source transcript.
- Plan labels, units, sequence and marks before writing.

Before submitting

- Check every module has been revised.
- Check diagrams, tables, formula symbols and units.
- Separate official syllabus information from personal elaboration.

REFERENCE VOCABULARY

Glossary

Study glossary

Term	Meaning in this lesson
Course Outcome	A capability the student should demonstrate after completing the course.
Module	An official syllabus unit with defined topics and hours.
CIE	Continuous Internal Evaluation.
ESE	End Semester Examination.
Source transcript	A preserved extract of the official syllabus used for coverage checking.

SOURCE-SUPPORTED REFERENCES

References and web resources

References listed in the official PDF

References are included in the official source PDF.

Web resources listed in the official PDF

- Sl.No Website Link <https://www.youtube.com/watch?v=CfCztL5DkTE>
<https://www.youtube.com/watch?v=qb3mvJ1gb9g>
<https://www.youtube.com/watch?v=QsLDBfApqol>
<https://www.youtube.com/watch?v=e5g2NmlUdck>
https://www.youtube.com/watch?v=GC6k_qbpwmA

IN-PAGE SEARCH**Search this lesson**

Prepared from the supplied official SITTTTR syllabus PDF for Course 5123A. Verify institutional updates against the current official curriculum.